

Project Summary: Employee Attrition and Performance Analysis

Project Objective: The primary goal was to develop an interactive Power BI dashboard to identify the root causes of employee turnover and analyze performance metrics to help HR management make data-driven decisions.

Analysis of Employee Retention and Workforce Performance Introduction:

This project involves building an advanced analytics platform using Power BI to study HR data.

The goal is to provide leadership with a clear view of why employees leave and how to help current staff succeed.

Problem Statement: High employee turnover (attrition) is a major cost for modern companies. By analyzing factors like job satisfaction, salary, and promotion history, we can predict and prevent future turnover.

Tools and Technologies: The primary tools used were Microsoft Power BI for visualization, Power Query for data transformation, and DAX for custom business logic.

Data Acquisition and Transformation

Milestone 1: Preparing Data for Advanced Analytics We got our hands on some raw HR data, which we imported from a couple of different file types - CSV and Excel. I made sure the data was clean by using Power Query. First, I removed any duplicate records.

Then, I checked that the data types were correct.

For instance, I made sure "Monthly Income" was treated as money and "Hire Date" was treated as a date. This helped keep everything consistent and accurate.

Feature Engineering: I created a Conditional Column for "Age Groups". This allowed for more meaningful demographics analysis by grouping individual ages into buckets like 18-25 and 26-35.

To tackle the issue of multiple tables, I decided to combine the Employee and Sheet1 tables.

This way, I could bring together all the relevant information, such as promotion dates and performance reviews, into one place.

Now, I have a single, unified view that makes it easier to create a useful dashboard. By merging these tables, I can see everything I need in one spot, which will help me make the dashboard more effective.

Data Modelling and Metric Development

Milestone 2:

Establishing Relationships and Business Logic Schema Design:

I defined relationships between different data tables to ensure the dashboard remained interactive.

Key Performance Indicators (KPIs): I developed several custom DAX measures to track the health of the organization.

Attrition Rate: A calculation showing the percentage of the workforce that has left.

Average Tenure: A measure of the average number of years employees stay with the company.

Promotion Gap: A formula calculating the average time since an employee's last promotion to identify potential "stagnation" areas.

Visualization and Dashboard Design

Milestone 3: Creating Interactive and Actionable Insights The main page of the Executive Dashboard gives you a quick overview, like a snapshot, of the key numbers.

It's got these KPI cards that show you the total number of employees and how many are leaving, which is really important for making big decisions.

Demographic Analysis: I used bar charts and donut charts to visualize turnover trends by gender, age, and marital status, helping HR identify high-risk groups.

Performance vs. Salary: I created a Scatter Plot to analyze the relationship between income levels and performance ratings.

This helps determine if higher-paid employees are meeting expectations or if low pay is causing top performers to leave.

I made it easier for users to focus on specific parts of the business by adding filters for departments and job roles.

This way, they can quickly look at the information that's most relevant to them.

Testing, Evaluation, and Conclusions

Final Testing and Strategic Recommendations I made sure that all the visuals show the right data when filtered by doing "filter checks" and manually checking the numbers.

This way, I could confirm that everything is accurate and working as it should.

Data Storytelling: From the dashboard, I identified key trends—for instance, noting that certain departments show higher attrition among employees who have longer promotion gaps.

In the end, this report is a useful tool for the people in charge of HR. It helps them tell a story with the data, which can then be used to reduce the number of employees who leave the company.

By looking at the gaps in how promotions are handled and how salaries are distributed, the company can make some changes to improve performance and keep employees from leaving.